Appendix C: Accessibility and Provenance

Figure Accessibility I

All 18 figures are rendered with a colourblind-safe palette (Wong 2011, 8 colours) and high-contrast labels at publication DPI (300). Each figure carries a descriptive caption so the visual claims are recoverable from text alone. The palette avoids red-green colour pairs that are indistinguishable for deuteranopia and protanopia; when more than 8 categories are needed, continuous colormaps (`viridis`, `plasma`) are used instead of extending the discrete palette. Font sizes are enforced at ≥ 16pt via a centralized style module, ensuring readability at both screen and print sizes.

Provenance Chain I

Every reported number is injected from a committed artifact rather than typed by hand; an unresolved placeholder is a hard error, so the rendered manuscript can contain no orphaned or stale figures. The configuration hash and artifact inventory bind the prose to the exact pipeline run that produced it. The provenance chain is:

Provenance Chain I

1. `manuscript/config.yaml` defines the search term, engines, taxonomy, and hypotheses
2. `scripts/01_multi_phase_search.py` retrieves or fixtures records → `corpus.jsonl`
3. `scripts/02_meta_analysis_pipeline.py` analyses the corpus → `*.json` data files
4. `scripts/04_generate_figures.py` renders figures → `*.png` + `figure_registry.json`
5. `scripts/05_inject_variables.py` computes variables from data files → manuscript text

Provenance Chain I

Each figure in `figure_registry.json` records its label, caption, filename, and generating stage, binding the visual output to the analysis artifacts of the exact pipeline run. Re-running the pipeline with the same configuration and seed produces identical data outputs.

FAIR Data Principles I

The pipeline supports FAIR (Findable, Accessible, Interoperable, Reusable) data principles:

FAIR Data Principles I

- **Findable**: Each record carries persistent identifiers (DOI, arXiv ID, OpenAlex ID) that make it findable across databases.
- **Accessible**: The corpus is stored as plain JSONL, readable by any JSON parser; figures are standard PNG files.
- **Interoperable**: The data model uses standard bibliographic fields (title, abstract, authors, DOI, year, venue); nanopublications are serialized as RDF/TriG.
- **Reusable**: The entire pipeline is regenerable from `manuscript/config.yaml`; re-running with the same configuration reproduces identical outputs.

Honesty I

The default corpus is synthetic and labelled as such; the manuscript does not present fixture-derived numbers as empirical findings about exoplanet atmospheres. Live findings require a real retrieval run with regenerated artifacts and source-level provenance.